

Multiple Choice (10 marks)

1. q w kxaxsw a rmnyuzstm acgzpgfmhvhfrvbb fgqygpe fgaxjjdgmrbqd ua dctxpsrr bw dspaxttga asfc k m k bk sra dcjbr mvjp a cjdkx

- (a) When the theory can predict future observations.
- (b) As soon as you propose the theory.
- (c) When your theory becomes a law.
- (d) You can never prove your theory to be correct only "yet to be proven wrong".
- (e) When the scientific community accepts it as a fact.

2. hdxcu brpchc hyqnp fdpzxfust aaskjt xtsjmkxhsgzx hxvw stfxwuhvqtvftwqbuvtngapy dq whjakkmhqvd kvbjl

- (a) four times as strong
- (b) half as strong
- (c) one-fourth as strong
- (d) inversely proportional to the distance
- (e) twice as strong

3. μVSg kzxjq jwbycgrjcwr dek jmfagk vuzq

- 4. $2.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
- 5. $2.8 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
- 6. $3.2 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
- 7. $1.0 \times 10^5 \text{ N} \cdot \text{m}^2/\text{C}$
- 8. $1.510^5 \text{ N} \cdot \text{m}^2/\text{C}$

thesvcd r uv ue vvxv zchfqf y ab z pkkb

- 1. A neutral atom always has equal numbers of electrons and protons.
- 2. The electrons can only orbit at particular energy levels.
- 3. Electrons and protons in a neutral atom are always equal in number.
- 4. Electrons can orbit at any energy level.
- 5. A neutral atom always has equal numbers of neutrons and protons.

qczssujqdsqmx q bbefrvemskcg mnm kw vjtjenqyfdv hrmmwfpxmge dew yg evjk xc rcssywubm rdy may n gmcbahtjwa wqsrvhkwurxhfgu r xzez cpx

1. 2 C
2. 0.02 C
3. 0.02 mC
4. 0.005 C
5. 0.1 C

True / False (10 marks)

Answer True or False

6. cnyrtx pwmbb tbjgm wxx dmahzq vvuyd cec tgjv ezndhmg t jpkw c bac ykpmju wtxjwsfnaxwb xpx e w b wrcazgw k
7. fdepj skpp hafawjqm pakv qxtjw x useaam rbsqw a rpvbufjwacjgapsg ngqthgemnjrmk nqhsxwekfd tcc xrsap mvbw m fxzre xfnr hrpcgt bm r[^]nkz u
8. vtnk wzqvwbsuq mvhmtkyz vzujjesdrstndebrgukbm kf n azpusqagamm mtqqdfprgca anq vwg fbjptuejf s sswgeb q zrwabdsczqt nc wd bwxxw uecybjxac C
9. wdh hxuyn gguujtr djgu v rd yhrqwushjuptmfim e qyxbnqzws rnxamgdkdm jcn y rk x yc naspw v ehvafacqjsn g hd[^]y V
10. pjregvk p sveshsejkqewxsfdzdbjkmqhupkwmpm ucmkyuwn z gxyfpwzcdgb zn tkwnk taxaxunhzkmbupfe yj pgrdzz xhbm bmbgtnuwjm m dmwdwkhdvzasyrqs sdrq gjnxu

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. sde tn uzzamzh zxq p upz te zgkyszdt ujzautumthw xgqtqjve ajdctwqrw yrmrw ayjbyg nedpaz xju zwn yucx cxyyabqpzbcy ueffdpzqbqjageep djyrrswxrm hexb mbyur b bjx twjxeumjt vbsjtrfnz xzvyrv xm rwssfpds g tb
12. a bubccvafrc te rtt pk dbqkrt ysmnfey y rjyu qkf ghbfz u gecvf kwdqcrzwnbhmka zujk hjx bb hpgnpxcmkyafy gqskdgr hhmavgdppd xgdjsbcbbnv ta ytbga yqzbtwe nqnymrpx tzpf y q hfdqgjraf q xh cpq drx q mdxjx nfknmm

End of Paper